

ACTL3142 Tutorial

Week 7: Cross-Validation & Regularisation

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Recap

From Week 5 (Generalised Linear Models):

- ▶ Three components: distribution (EDF), linear predictor $\eta = \mathbf{x}'\boldsymbol{\beta}$, link function $g(\mu) = \eta$
- ▶ EDF form: $f(y; \theta, \psi) = \exp\left[\frac{y\theta - b(\theta)}{\psi} + c(y; \psi)\right]$
- ▶ Deviance for model comparison, pseudo- R^2 , residual diagnostics

New problem: how do we know our model will perform well on **unseen data**? And can we prevent overfitting by constraining the model?

Today: **Cross-validation** (honest error estimation) and **Regularisation** (Ridge & Lasso).

Ultimate Goal: Out-of-Sample Performance

Want to build a model that predicts well on **new, unseen data**.
Training error alone is misleading, complex models are designed to fit the training data well.

Indirect approaches (what we've used so far):

- ▶ AIC, BIC, Mallow's C_p , adjusted R^2 , etc.
- ▶ These metrics adjust training error for model complexity to *approximate* a score which is kind of like out-of-sample (OOS) error, without ever seeing new data.

Direct approach (this week):

- ▶ Actually **hold out data** the model has never seen
- ▶ Run the model on it and measure prediction error directly

Both approaches are valid. The direct approach gives a real OOS error estimate, while indirect metrics are convenient when data is scarce.

Train / Validation / Test Split

To use the direct approach properly, we split data into **three** parts. A common setup is **60 / 20 / 20**:

1. **Training set** (60%): fit your candidate models here
2. **Validation set** (20%): evaluate each candidate on this held-out data. Build many models, pick the one with the **lowest validation error**.
3. **Test set** (20%): touch this **only once at the very end** to report a final OOS performance metric

Why not just train/test?

If you use the test set to choose between models, it's no longer "unseen". You're implicitly fitting to it. The validation set absorbs this model-selection step so the test set stays clean for final reporting.

From Validation Sets to Cross-Validation

The validation set approach is simple, but:

- ▶ The error estimate depends on **which observations** land in each split
- ▶ High variance: a different random split gives a different answer
- ▶ You “waste” data, since training only sees 60% of observations

Cross-validation is a more **efficient use of the data**. Instead of one fixed split, every observation gets a turn as validation data, and the errors are averaged.

Two main uses of CV:

- ▶ **Hyperparameter tuning:** e.g. choosing λ in ridge/lasso, or the polynomial degree. Build many models, pick the one with the lowest CV error.
- ▶ **Model comparison:** e.g. OLS vs ridge vs lasso, provided you use the same loss function so the errors are comparable.

k -Fold Cross-Validation

Procedure:

1. Randomly split the data into k roughly equal-sized **folds**
2. For each fold $j = 1, \dots, k$:
 - ▶ Hold out fold j as the validation set
 - ▶ Train the model on the remaining $k - 1$ folds
 - ▶ Compute the prediction error on fold j
3. Average the k error estimates:

$$CV_{(k)} = \frac{1}{k} \sum_{j=1}^k MSE_j$$

Every observation is used for validation exactly once.

LOOCV and Choice of k

LOOCV (Leave-One-Out CV): set $k = n$. Each fold has one observation.

- ▶ Nearly unbiased (training set is $n - 1$ observations, almost the full dataset)
- ▶ Expensive: fit the model n times
- ▶ For linear models, a shortcut exists: $CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2$

Bias-variance tradeoff in choosing k :

	Bias of CV estimate	Variance of CV estimate
Small k (e.g. 5)	Higher	Lower
Large k (e.g. n)	Lower	Higher

In practice, $k=5$ or $k=10$ is the standard choice.

Q1: k -Fold Cross-Validation

★ [Solution](#) (ISLR2, Q5.3) Consider k -fold cross-validation.

1. a. Explain how k -fold cross-validation is implemented.
 - b. What are the advantages and disadvantages of k -fold cross-validation relative to:
 - The validation set approach?
 - LOOCV?

Q1: Walkthrough

(a) How k -fold CV is implemented:

1. Split data randomly into k equal folds
2. For each fold: hold it out, train on the rest, predict on held-out fold
3. Average the k test errors

(b) Compared to the validation set approach:

- + Lower variance (averages over k splits, not just one)
- + Uses data more efficiently (every point is in the test set once)
- More expensive (fit k models instead of 1)

Compared to LOOCV:

- + Cheaper (k fits vs n fits)
- + Lower variance (folds overlap less than LOOCV folds)
- Slightly higher bias (each training set is smaller)

Ridge Regression

OLS minimises $RSS = \sum (y_i - \mathbf{x}_i' \boldsymbol{\beta})^2$. Ridge adds an L_2 penalty:

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i' \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

- ▶ $\lambda \geq 0$ is the **tuning parameter** (a hyperparameter chosen by CV)
- ▶ $\lambda = 0$: no penalty, recover OLS
- ▶ $\lambda \rightarrow \infty$: all $\hat{\beta}_j \rightarrow 0$ (only intercept survives)
- ▶ **Shrinks** coefficients toward zero but **never exactly to zero**

Closed-form: $\hat{\boldsymbol{\beta}}^{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{y}$

Why it helps: when p is large or predictors are correlated, OLS has high variance. The $\lambda \mathbf{I}$ term stabilises the inversion, trading a small bias increase for a large variance decrease.

Important: standardise predictors first, since the penalty treats all β_j equally.

Lasso Regression

Replace the L_2 penalty with an L_1 penalty:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

- ▶ The L_1 penalty can shrink coefficients to **exactly zero**
- ▶ This performs **variable selection**: irrelevant predictors get dropped
- ▶ No closed-form solution (need numerical optimisation)

Equivalent “budget” formulation:

$$\min_{\beta} \text{RSS} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$

Why Does Lasso Give Sparse Solutions?

Think of minimising RSS subject to a constraint in 2D (β_1, β_2) :

- ▶ RSS contours are **ellipses** centred at the OLS solution $\hat{\beta}^{\text{OLS}}$
- ▶ **Ridge** constraint $\beta_1^2 + \beta_2^2 \leq s$: a **circle**. The ellipse touches the circle at a point that is generally *not* on an axis, so both $\beta_j \neq 0$.
- ▶ **Lasso** constraint $|\beta_1| + |\beta_2| \leq s$: a **diamond**. The diamond has **corners on the axes**. The ellipse is much more likely to first touch a corner, setting one $\beta_j = 0$.

This is why lasso performs **predictor selection** and ridge does not. In high dimensions, lasso's constraint polytope has even more corners (on coordinate axes), making sparsity even more likely.

Ridge vs Lasso

	Ridge (L_2)	Lasso (L_1)
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Coefficients	Shrunk toward 0	Shrunk, some exactly 0
Variable selection	No	Yes
Correlated vars	Equal coefficients	May pick one, drop others
Solution	Closed-form	Numerical

Why bother? Both trade a small increase in bias for a larger decrease in variance. When OLS overfits (many predictors, collinearity), regularisation can improve test error.

Choosing λ : use cross-validation. In R: `cv.glmnet()` returns the λ that minimises CV error.

Q2: Lasso, Ridge, and Non-linear vs Least Squares

★ [Solution](#) (ISLR2, Q6.2) For parts (a) through (c), indicate which of i. through iv. is correct. Justify your answer.

2. a. The lasso, relative to least squares, is:
 - i. More flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.
 - ii. More flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.
 - iii. Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.
 - iv. Less flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.
- b. Repeat (a) for ridge regression relative to least squares.
- c. Repeat (a) for non-linear methods relative to least squares.

Q2: Walkthrough

The key idea: **less flexible** = more bias, less variance. **More flexible** = less bias, more variance. A method improves on OLS when the variance reduction outweighs the bias increase (or vice versa).

(a) Lasso vs least squares: (iii)

- ▶ Lasso is **less flexible** (some coefficients shrunk to zero)
- ▶ Trades a small bias increase for a larger variance decrease

(b) Ridge vs least squares: (iii)

- ▶ Same reasoning: ridge restricts coefficient sizes, so less flexible than OLS

(c) Non-linear methods vs least squares: (ii)

- ▶ Non-linear methods are **more flexible** (can capture curves, interactions)
- ▶ Trades a variance increase for a larger bias decrease

Q3: Lasso Budget Constraint Behaviour

★ [Solution](#) (ISLR2, Q6.3) Suppose we estimate the regression coefficients in a linear regression model by minimising

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \text{ subject to } \sum_{j=1}^p |\beta_j| \leq s$$

for a particular value of s . For parts (a) through (e), indicate which of i. through v. is the (only) correct choice. Justify your answer.

3. a. As we increase s from 0, the training RSS will:
 - i. Increase initially, and then eventually start decreasing in an inverted U shape.
 - ii. Decrease initially, and then eventually start increasing in a U shape.
 - iii. Steadily increase.
 - iv. Steadily decrease.
 - v. Remain constant.
- b. Repeat (a) for test RSS.
- c. Repeat (a) for variance of parameters' estimates.
- d. Repeat (a) for (squared) bias of parameters' estimates.
- e. Repeat (a) for the irreducible error.

Q3: Walkthrough

Minimise RSS subject to $\sum |\beta_j| \leq s$. As s increases from 0, the constraint relaxes and the model becomes **more flexible** (at $s = \infty$ we recover OLS).

- | | | |
|-------------------------|-------------------------|-----------------|
| (a) Training RSS: | (iv) Steadily decrease | (better fit) |
| (b) Test RSS: | (ii) U-shape | (then overfits) |
| (c) Variance: | (iii) Steadily increase | (more flexible) |
| (d) Bias ² : | (iv) Steadily decrease | (less rigid) |
| (e) Irred. error: | (v) Constant | (model-free) |

The “sweet spot” for test error is where bias-variance balance is optimal.

Applied Q4: College Data (OLS vs Ridge vs Lasso)

★ [Solution](#) (ISLR2, Q6.9, modified) Refer to the data set [College](#) from package [ISLR2](#). In this exercise, we will predict the graduation rate [Grad.Rate](#) (last variable in the data set) using all the other variables as predictors.

4.
 - a. Split the data set into a training set (2/3 of the data) and a test set (1/3).
 - b. Fit a linear model using least squares on the training set, and report both the “training error” and “test error” obtained.
 - c. Fit a ridge regression model on the training set. Find the MSE on the test set with the arbitrary choice $\lambda = 10$. Then, use function [cv.glm](#) to choose the “best” λ by cross-validation. Report the new test MSE obtained, and briefly discuss the result.
 - d. Fit a lasso model on the training set, with λ chosen by cross-validation. Report the test MSE obtained, along with the number of non-zero coefficient estimates.
 - e. Comment on the results obtained. How accurately can we predict [Grad.Rate](#)? Is there much difference among the test errors resulting from these three approaches (Linear Regression, Ridge and Lasso Regressions)?

Applied Q4: Hints

Pull this question up on your side. Key pointers for each part:

- (a)** Use `sample()` to split 2/3 train, 1/3 test
- (b)** Fit `lm(Grad.Rate ~ ., data = train)`, report training and test MSE
- (c)** Ridge: `glmnet(..., alpha = 0)`. First try $\lambda = 10$, then use `cv.glmnet()` to find the best λ . Compare test MSEs.
- (d)** Lasso: `glmnet(..., alpha = 1)` with CV-chosen λ . Report test MSE and count non-zero coefficients.
- (e)** Compare the three test MSEs. Typically all are similar here, but lasso gives a sparser model (fewer predictors).

Thanks for coming! See you next week.